

A counterexample to Conjecture 00000007680

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The conjecture explicitly permits positive integers a, b, m and asserts that, for every admissible modulus m , the number of valid residues b in $\{0, \dots, a - 1\}$ is the number $\omega(m)$ of distinct prime factors of m . The exact-count assertion already fails at $a = 2, m = 1$.

Let $p(n)$ denote the ordinary partition function. Every integer is congruent to zero modulo 1. Consequently

$$p(2n + b) \equiv 0 \pmod{1} \quad (n \geq 0)$$

holds for both $b = 0$ and $b = 1$. Thus the number of valid residues in the stated set $\{0, 1\}$ is 2, whereas $\omega(1) = 0$. If the opening requirement that b be positive is also applied to the residue count, the allowed set instead contains the single valid residue $b = 1$; its count is 1, still different from 0. In particular $(a, b, m) = (2, 1, 1)$ satisfies every positivity condition.

The first assertion about the prime factors of m does not exclude this example: 1 has no prime factors, so any condition required of all its prime factors holds vacuously. The example uses the written positive-modulus domain. A reformulation imposing $m \geq 2$ would be a different assertion and is not addressed here.

Formal verification. The Lean source defines the usual prime predicate, distinct-prime-factor count by a finite list of the possible divisors, and the congruence with a universal quantifier over every $n \in \mathbb{N}$. Residue counts filter the entire type $\text{Fin}(a)$; they are not counts of a selected sample. The theorem `conjecture7680_counterexample` disproves the exact-count clause under both readings of the positivity of b .

Its parameter $p : \mathbb{N} \rightarrow \mathbb{N}$ is arbitrary. This strengthens the result: it holds for every sequence, hence directly for the actual partition function, without replacing partitions by a finite table or assuming any unproved identity for their values. The statement requires the modulus to have a valid positive residue before applying the purported count; the formal witness is $b = 1$. Lean 4.19.0 checks the proof using only `Std`. There are no admitted lemmas, custom axioms, or native evaluation proofs.